

The Simplex Method

• Matrix notation

For the sake of easy expression, we introduce the following matrix notation.

Partition A as $A = [B \ N]$ and partition x as $x = \begin{bmatrix} x_B \\ x_N \end{bmatrix}$. B is the current basis matrix and thus consists of the basic columns. N is a submatrix of A formed by the nonbasic columns. x_B and x_N are corresponding basic and nonbasic variables respectively. Then $Ax = b$ can be written as

$$Bx_B + Nx_N = b$$

The canonical form corresponding to basis B can be represented as

$$I_m x_B + B^{-1}N x_N = B^{-1}b.$$

Remark: WLOG, we assume the first m variables are basic variables in our deduction (if not claimed otherwise).

That is, we assume $B = [A_1 \ A_2 \ \dots \ A_m]$.

• Reduced cost

Now let's take the objective function into consideration

Let $A = [B \ N]$, $x = \begin{bmatrix} x_B \\ x_N \end{bmatrix}$, and $c = \begin{bmatrix} c_B \\ c_N \end{bmatrix}$. Then

$$Ax = b \Leftrightarrow Bx_B + Nx_N = b \Leftrightarrow x_B = B^{-1}b - B^{-1}Nx_N$$

$$\begin{aligned} z = C^T x &= C_B^T x_B + C_N^T x_N = C_B^T (B^{-1}b - B^{-1}N x_N) + C_N^T x_N \\ &= C_B^T B^{-1}b - (C_B^T B^{-1}N - C_N^T) x_N \\ &= C_B^T B^{-1}b - \underbrace{\sum_{j=m+1}^n (C_B^T B^{-1}A_j - c_j)}_{\bar{c}_j} x_j \quad (*) \end{aligned}$$

Def The reduced cost of a variable x_j wrt a basis matrix B is defined as $\bar{c}_j = C_B^T B^{-1}A_j - c_j$

eg

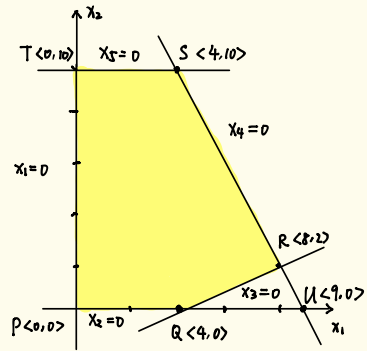
$$\max Z = 2x_1 + 3x_2$$

$$\text{s.t. } x_1 - 2x_2 + x_3 = 4$$

$$2x_1 + x_2 + x_4 = 18$$

$$x_2 + x_5 = 10$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$



We see that $C = [2 \ 3 \ 0 \ 0 \ 0]^T$. We consider two examples.

$$\textcircled{1} B = [A_3 \ A_4 \ A_2] = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_B^T = [C_3 \ C_4 \ C_2] = [0 \ 0 \ 3]$$

$$\begin{aligned} \bar{C}_1 &= C_B^T B^{-1} A_1 - C_1 = [0 \ 0 \ 3] \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - 2 \\ &= [0 \ 0 \ 3]^T \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - 2 \end{aligned}$$

$$= 0 - 2 = -2$$

$$\bar{C}_2 = C_B^T B^{-1} A_2 - C_2 = [0 \ 0 \ 3]^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - 3 = 0$$

$$\bar{C}_3 = C_B^T B^{-1} A_3 - C_3 = [0 \ 0 \ 3]^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 0 = 0$$

$$\bar{C}_4 = C_B^T B^{-1} A_4 - C_4 = [0 \ 0 \ 3]^T \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - 0 = 0$$

	x_1	x_2	x_3	x_4	x_5	RHS
x_3	1	0	1	0	2	24
x_4	2	0	0	1	-1	8
x_2	0	1	0	0	1	10

$B^{-1}A_1$

$$\begin{aligned} \bar{C}_5 &= C_B^T B^{-1} A_5 - C_5 \\ &= [0 \ 0 \ 3]^T \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} - 0 \\ &= 3 \end{aligned}$$

Prop The reduced cost of any basic variable is 0.

e.g. (cont'd)

$$\textcircled{2} \quad B = [A_3 \ A_4 \ A_5] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\bar{C}_1 = C_B^T B^{-1} A_1 - C_1 = [0 \ 0 \ 0]^T \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - 2 = -2$$

$$\bar{C}_2 = C_B^T B^{-1} A_2 - C_2 = [0 \ 0 \ 0]^T \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} - 3 = -3$$

$$\bar{C}_3 = \bar{C}_4 = \bar{C}_5 = 0$$

	x_1	x_2	x_3	x_4	x_5	RHS
x_3	1	-2	1	0	0	4
x_4	2	1	0	1	0	18
x_5	0	1	0	0	1	10

• Optimality condition

From equation (*),

$$Z = C_B^T B^{-1} b - \sum_{j=m+1}^n \underbrace{(C_B^T B^{-1} A_j - C_j)}_{\bar{C}_j} x_j$$

we see that if $\bar{C}_j \geq 0$ for all nonbasic variables x_j ($j = m+1, \dots, n$), then increasing the value of x_j from 0 to a positive number cannot improve the objective value. Since $x_j \geq 0$ ($j = m+1, \dots, n$) for all feasible solutions, we have the following optimality condition.

Thm: A basic feasible solution is optimal if $\bar{C}_j \geq 0$ for all nonbasic variables x_j .

- The simplex tableau.

We add the reduced costs to the previous tableau to get the simplex tableau.

	z	x	RHS	
z	1	$C_B^T B^{-1} A - C^T$	$C_B^T B^{-1} b$	(Row 0)
x_B	$\vec{0}$	$B^{-1} A$	$B^{-1} b$	(Row 1 - Row m)

eg. The simplex tableaus of the previous example.

	z	x_1	x_2	x_3	x_4	x_5	RHS
z	1	-2	0	0	0	3	30
x_3	0	1	0	1	0	2	24
x_4	0	2	0	0	1	-1	8
x_2	0	0	1	0	0	①	10

R_0 means $z - 2x_1 + 3x_5 = 30$

R_2 means $2x_1 + x_4 - x_5 = 8$

basic variables: x_3, x_4, x_2

BFS: $x = (0, 10, 24, 8, 0)$

objective value: $z = 30$

	z	x_1	x_2	x_3	x_4	x_5	RHS
z	1	-2	-3	0	0	0	0
x_3	0	1	-2	1	0	0	4
x_4	0	2	1	0	1	0	18
x_5	0	0	1	0	0	1	10

R_0 means $z - 2x_1 - 3x_2 = 0$

R_2 means $2x_1 + x_2 + x_4 = 18$

basic variables: x_3, x_4, x_5

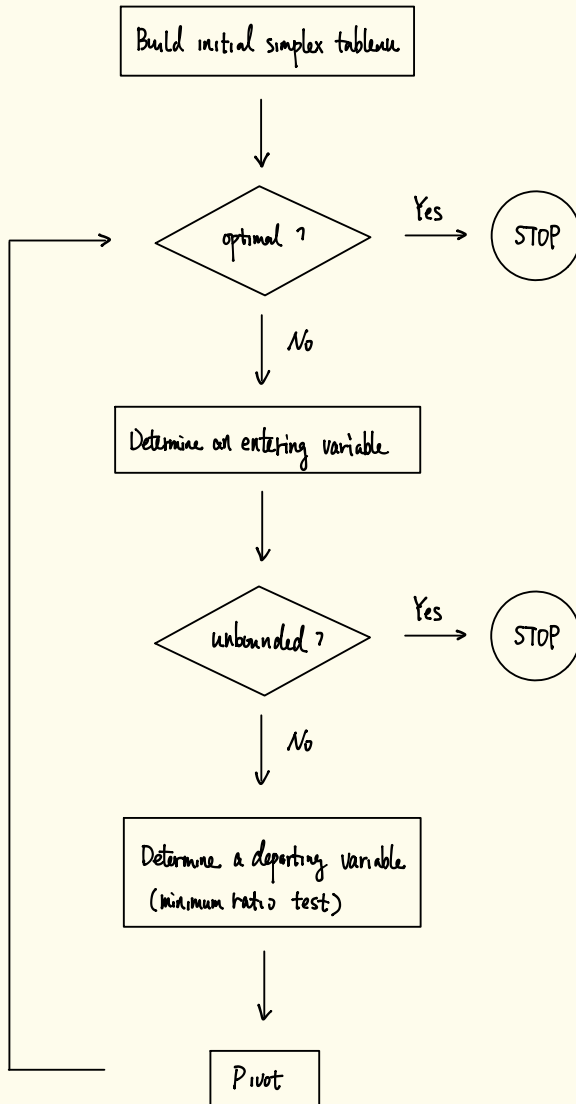
BFS: $x = (0, 0, 4, 18, 10)$

objective value: $z = 0$

New reduced costs need to be calculated when the basis is changed. Fortunately, the reduced costs (Row 0) can be updated at the same time in the process of pivoting.

Exercise: Pivot about the circled element in the left tableau to get the tableau on the right.

- The Simplex Method



- Determine an entering variable

In principle, we can choose any nonbasic variable x_k with a negative reduced cost $\bar{c}_k$ as the entering variable

Usually, people choose x_k with the most negative reduced cost to achieve fastest ascent. (Dantzig's rule)

e.g.,

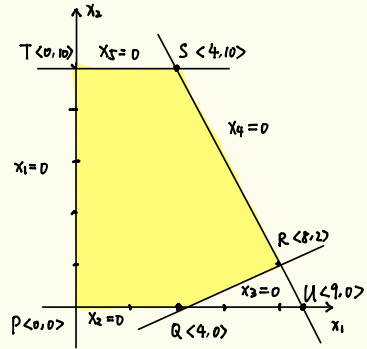
$$\max \quad z = 2x_1 + 3x_2$$

$$\text{s.t.} \quad x_1 - 2x_2 + x_3 = 4$$

$$2x_1 + x_2 + x_4 = 18$$

$$x_2 + x_5 = 10$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$



	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	1	-2	-3	0	0	0	0
x_3	0	1	-2	1	0	0	4
x_4	0	2	1	0	1	0	18
x_5	0	0	①	0	0	1	10

$\bar{c}_2 = -3$ is the most negative reduced cost. Let x_2 enter the basis.

Minimum ratio test, $\frac{18}{1} > \frac{10}{1}$. Kick x_5 out of the basis.

Pivot.

	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	1	-2	0	0	0	3	30
x_3	0	1	0	1	0	2	24
x_4	0	②	0	0	1	-1	8
x_2	0	0	1	0	0	1	10

	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	1	0	0	0	1	2	38
x_3	0	0	0	1	$-\frac{1}{2}$	$\frac{5}{2}$	20
x_1	0	1	0	0	$\frac{1}{2}$	$-\frac{1}{2}$	4
x_2	0	0	1	0	0	1	10

optimal solution: $x^* = (4, 10, 20, 0, 0)$

optimal value: $z^* = 38$

• The unbounded case

In the current tableau, if there exists a nonbasic variable x_k such that $\bar{c}_k < 0$ and $\alpha_{ik} \leq 0$ (i.e., the column is nonpositive and the reduced cost is negative), then the problem is unbounded.

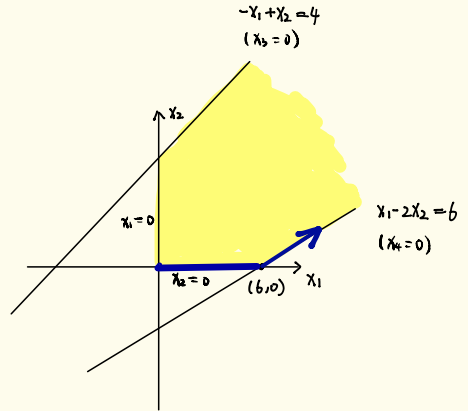
eg

$$\max Z = 5x_1 + 3x_2$$

$$\text{s.t. } -x_1 + x_2 + x_3 = 4$$

$$x_1 - 2x_2 + x_4 = 6$$

$$x_1, x_4 \geq 0$$



	z	x_1	x_2	x_3	x_4	RHS
z	1	-5	-3	0	0	0
x_3	0	-1	1	1	0	4
x_4	0	①	-2	0	1	6

The objective value can be arbitrarily large along this ray.

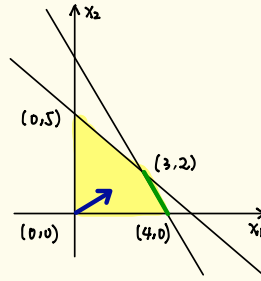
→

	z	x_1	x_2	x_3	x_4	RHS
z	1	0	-13	0	5	30
x_3	0	0	-1	1	1	10
x_1	0	1	-2	0	1	6

Unbounded

• Alternative optimal solution

eg $\max Z = 2x_1 + x_2$
 $\text{st } 2x_1 + x_2 + x_3 = 8$
 $x_1 + x_2 + x_4 = 5$
 $x_1, x_2, x_3, x_4 \geq 0$



	Z	x_1	x_2	x_3	x_4	RHS
Z	1	-2	-1	0	0	0
x_3	0	2	1	1	0	8
x_4	0	1	1	0	1	5

→

	Z	x_1	x_2	x_3	x_4	RHS
Z	1	0	0	1	0	8
x_1	0	1	$\frac{1}{2}$	$\frac{1}{2}$	0	4
x_4	0	0	$\frac{1}{2}$	$-\frac{1}{2}$	1	1

Optimal

Further analysis

Note that $\bar{C}_2 = 0$ Increasing x_2 won't change the objective value.

	Z	x_1	x_2	x_3	x_4	RHS
Z	1	0	0	1	0	8
x_1	0	1	$\frac{1}{2}$	$\frac{1}{2}$	0	4
x_4	0	0	$\frac{1}{2}$	$-\frac{1}{2}$	1	1

→

	Z	x_1	x_2	x_3	x_4	RHS
Z	1	0	0	1	0	8
x_1	0	1	0	1	-1	3
x_2	0	0	1	-1	2	2

$$x^* = (4, 0, 0, 1)$$

$$x^{**} = (3, 2, 0, 0)$$

Both are optimal BFS.